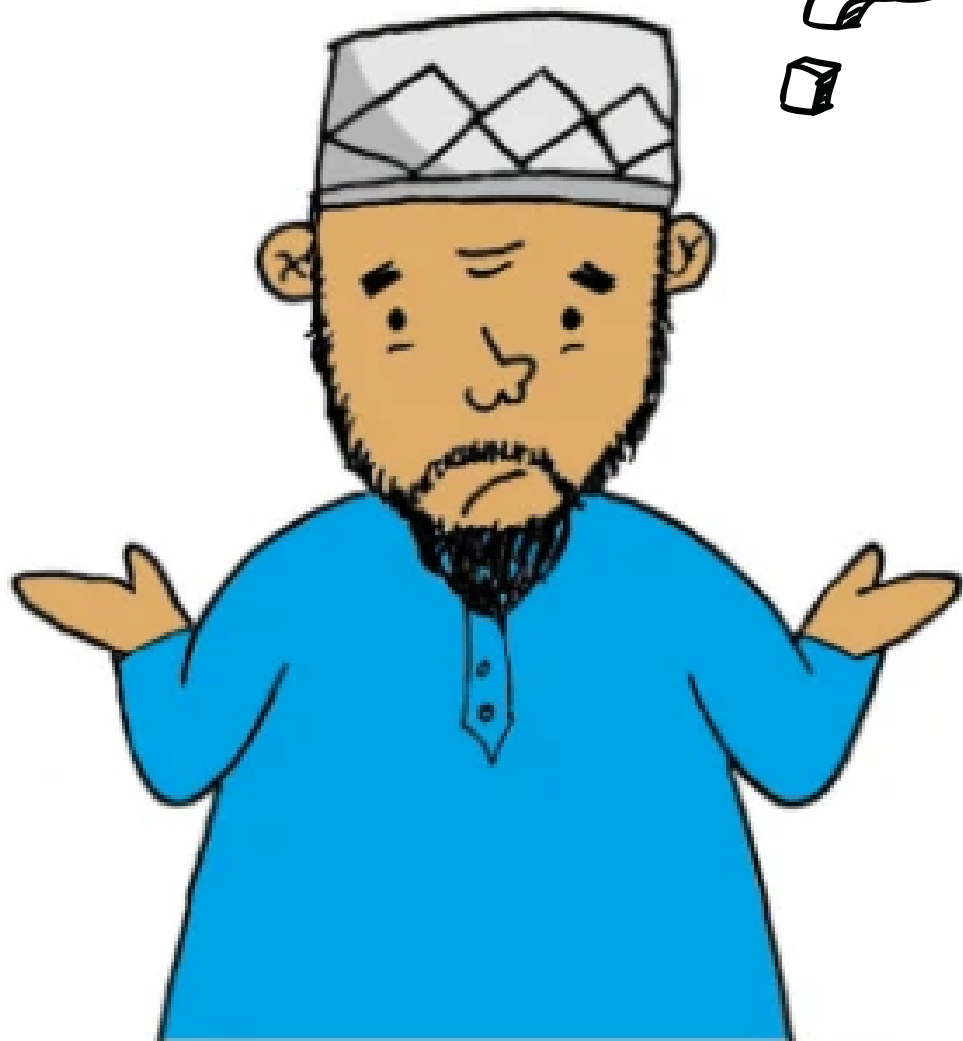




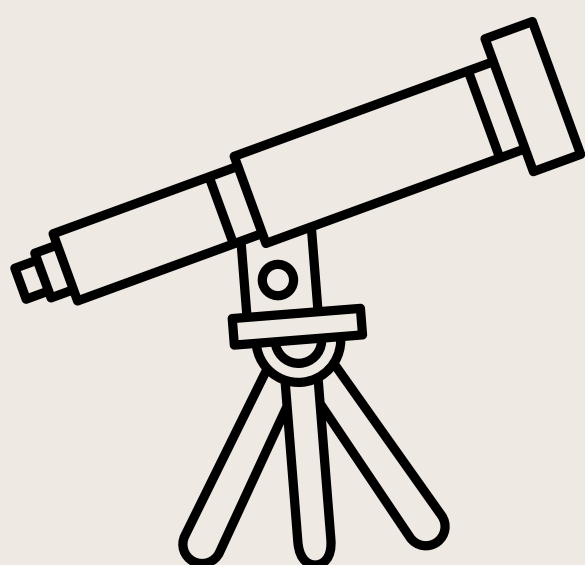
DI MANAKAH KAWANKU?

BY PEMBANTU TOK BAH



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INTRODUCTION

“Di Manakah Kawanku” is a 3D based mobile game application in which to help the main character, Tok Bah, to find his friends (animals in farm) that are located in the maze according to the story flow. Players have to find a total of 9 different shadows of animals and Tok Bah’s grandson in the maze. To continue the game, they must answer the quiz, which is the name of the shadowed animal each time they meet the underbrush. The players will need to search for different animals each time.



PROBLEM STATEMENT

- In this era of science and technology, we found that younger generations lack interest in reading story books.
- This is because there are many high-tech products such as smartphones, tablets, computers, and laptops, and children nowadays prefer to spend their free time on these products compared to reading books.

METHOD PROJECT

- The objective of the project is to promote the interest of the younger generation to read.
- Our project aims to combine the story books’ contents and game to attract the younger generation to get interested in reading the story books through the game.

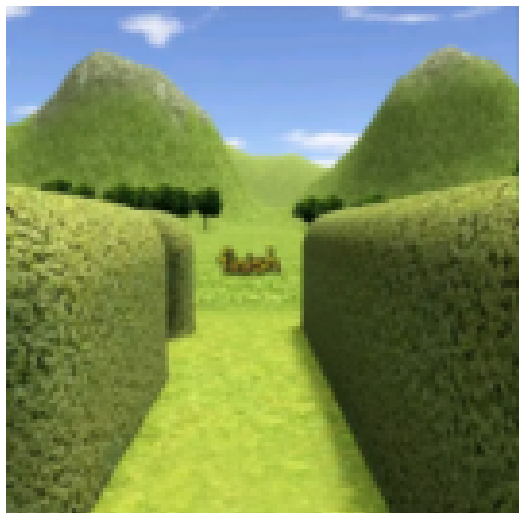
PROJECT USABILITY

- In this era of science and technology, we found that younger generations lack interest in reading story books.
- According to the Kids and Family Reading Report by Scholastic, only 40% of 8-years-old say that they love reading, and even worse for 9-years-old, which are 28%.
- Therefore, our target audiences are the children between 6 to 12 years old.



COMPETITIVE ANALYSIS

Below shows the comparison between the three chosen similar games.

Application Name	3D Maze / Labyrinth	Labo: 3D Maze	3D Maze Level 100
Icon			
Developer	Mobadu	RHM Interactive OÜ	Wasabi Applications
Published in Year	5 May 2014	25 Jun 2016	25 Dec 2014
Key features	- Contain 12 different mazes - Mini map not provided, but players can observe the map before entering. - Players need to exit from the maze within 10 minutes	- Contain 24 different mazes - Mini map provided - Players need to collect enough keys to open the door before exit from the maze. - 3 lives. Lives will be deducted if players fall into a trap (rolling stone, tornado, floor spikes, fire...)	- Contain 170 different mazes - Mini map provided, but will only show the part that player steps before. - Players need to collect the correct color of key before exit from the maze as soon as possible. - Escape faster, players will get more stars.
Performance	- Overall performance is average	- Overall performance is good.	- Overall performance is good.
Gameplay	- Third-person view - 3D maze-escape game - Horizontal screen	- Third-person view - 3D maze adventure escape game - Horizontal screen	- First-person view - 3D maze-escape game - Vertical screen
Metaphor	- Font type: similar with Nkotb Fever - Inconsistent design of icon - Map indicates the show the right path - Timer	- Font type: Black Note - Consistent design of icon: yellow + brown - Heart indicates live - Key indicates the requirement to pass the level - Tornado, rolling stone, floor spikes and fire indicate danger	- Font type: Arial - Consistent design of icon: white + transparent - Stars indicates performance of the players - Timer
Language	English	English	English
Strength	- Warn players and ask for confirmation when players choose to leave the game - Players can unlock new skin if they finish certain levels. - Players can view their highscores at main menu	- Detailed 3D model - Joyful background music and sound effect - Aesthetic design of interface - Attractive and challenging game mechanics - Players can choose to play as 5 different	- Provide many levels. Players need to play for a longer time to complete the whole progress. - Time killer game. Average time to complete a level is very short. - Easy to operate. The

	- Easy to understand even if there is no instruction.	- characters - Free to rotate viewpoint in game	buttons are large
Weakness	- Many advertisement - Limited view perspective (can only turn 180°) - Too easy to escape. Even if players do not memorize the route, they can jump to view the whole map - Reverse the position of ‘Jump’ and Movement buttons - Setting can only be adjusted at the main menu	- Does not provides instruction - Does not have any tutorial or introduction about its element	- Poor navigation. It does not have any tutorial or introduction about its element (tools, teleport, rules) - Limited view perspective (only rotate 90° per one tap) - Setting can only be adjusted at the main menu

3D Maze/Labyrinth

- Visibility: - Good. All controls are placed in highly visible locations.
- Feedback: - Poor. Reactions are fast, but loading screen does not show current loading progression, buttons does not have pushdown effect and sound effect.
- Constraints: - Poor. Settings are allowed to adjust at the main menu only, visions are limited at 180°.
- Mapping: - Poor. Usually, games have a movement key at left and jump button at right, but this game is the other way round.
- Consistency: - Poor. The icons do not have a consistent design, some are black and filled, some have black outline and transparent in the middle.
- Affordances: - Good. Players can understand every object easily.



Labo: 3D Maze

- Visibility: - Good. All controls are placed in highly visible locations.
- Feedback: - Good. Reactions are fast, there is pushdown effect and sound effect when a button is pressed, loading screen shows current loading progression.
- Constraints: - Average. Players are free to adjust point of view, but sometimes will cause clipping.
- Mapping: - Good. The mapping is natural and convenient for players.
- Consistency: - Good. The same styles remained throughout the game.
- Affordances: - Average. No instruction on how to play, but players can understand everything in the first few levels.

3D Maze Level 100

- Visibility: - Good. All controls are placed in highly visible locations.
- Feedback: - Average. Reactions are fast, and there is a vibration screen effect when players are knocking the wall, but buttons do not have pushdown effect but sound effect, loading screen does not show current loading progression
- Constraints: - Average. Limited point of view reduces the possibility of bugs, but limits the player's experience too.
- Mapping: - Good. The mapping is natural and convenient for players
- Consistency: - Good. The same styles are remained throughout the game
- Affordances: - Poor. No instruction on how to play, and it is hard to understand what are the functions of the items, where the teleport will be located, what is the color of the key that needed to be found.

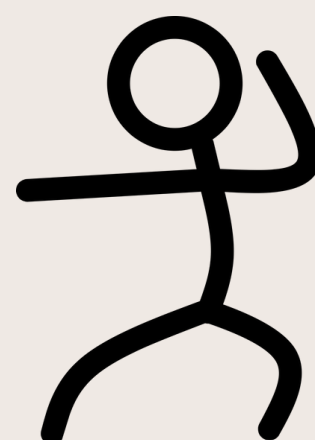


DESIGN THINKING PROCESS

The story is that Tok Bah saw that it was his grandson's birthday and wanted to gather all his friends from the farm to celebrate his grandson's birthday, so we set off on this journey to find them. The story ends with Tok Bah getting all his farm friends together to celebrate with his grandson, and it's a very happy ending, followed by a score and time calculation.

Firstly, our initial idea was that we would use the map in the storybook to allow the user to follow the map to the tok bah's friends and exits, thus winning the game. After some group discussions, we found that we wanted to make the game more enjoyable for the users, so we optimized the game into a 3D maze with some bushes to hide Tok Bah's friend and his grandson, Anas. Afterward, we set the user as Tok Bah's character, and also in the first person view.

Once the user has found the bush, a small window will pop up which will allow the user to interact with our game in a small way. This game is to drag and drop with the provided questions and answers. If the users get it right, they will get a corresponding animal sound, if they get it wrong, they will have to answer again. In order to take into account the difficulty of the game, we have set up two levels, the first is the easy mode with a map for reference, and the other is the hard mode, which does not have a map for reference and has a time calculation, so that users can keep breaking through to set their highest time record.



CRITICAL THINKING

Some requirements that might be needed to be implemented in the task:

- The need for a consistency of button display pattern, instead of using multiple patterns of button for clicking the correct instruction.
- The need for a happy and pleasing background music throughout the application, but not annoying music.
- The need for harmonizing and suitable colors for user age between 6-12 years old.
- The need for an easily accessible mechanism for a time limit 3D maze interactive mobile application.

REQUIREMENT ANALYSIS

Functional Requirements

Interface requirements

Throughout the application, the application shall be able to implement all the multimedia elements such as text, audio and animation in the application

Throughout the application, the application will be able to function in offline-based.

Throughout the application, the application will be able to fully support in 2D and 3D animated graphics.

User interaction

While playing games, the application shall be able to let users brain training throughout the application.

While playing the games, the application will be able to increase the difficulties of the level of games applications.

Autonomous system activities

While complete a level, the application automatically end the game and unlock the new next level.

When users answer wrong question, the application will deduct one life.

At the end of the game, the application will show the final score obtained by the user.

3.3.2 Non-Functional Requirements

No	Requirements	Elaboration
1	Usability	Users shall be allowed to access this interactive application anytime and anywhere.
2	Performance	This application is provided as an offline-mode-based interactive application.
3	Legal	Users shall be allowed to view the information that contains in the mobile application but cannot modify the data.
4	Implementation	This application shall be able to operate by using any android mobile-foundation platform except iOS-based mobile system.
5	Cultural	This application shall be using Bahasa Melayu as a medium language which is easy for the Malaysian public to understand.

USER MANUAL

Information Architecture

Main Menu
Content: - Start button - Score button - Setting
Action: - User can start the story - User can adjust the setting - User can see the score

Loading
Content: - Loading bar - Illustration
Action: User can see the loading screen with illustration

Story
Content: - Story Scenes - Skip button
Action: - User can watch and read the story - User can skip the story

In Game
Content: - Joystick - Map (in Easy mode) - Pause button - Timer - Gameplay Interface
Action: - User can move the character by using joystick - User can pause or left the game - User can see the time left - User can play with map

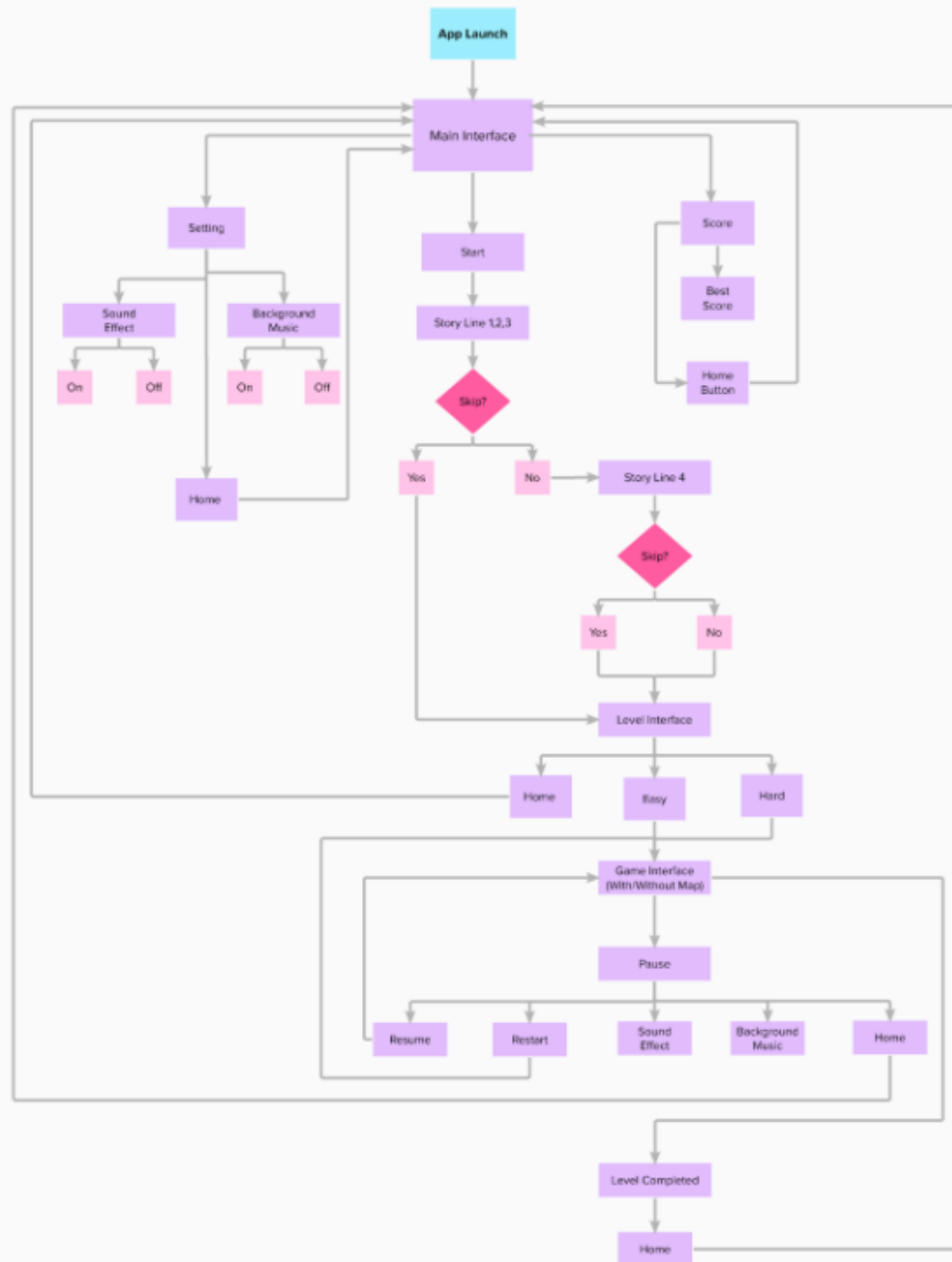
Score
Content: - Result - Home button
Action: - User can see the result of both Easy mode and Hard mode - User can return to Main Menu by clicking Home button

Setting
Content: - Controller setting - Sound effect switch - Music switch - Home button
Action: - User can choose the preferred controller they want - User can switch on or off for Music and Sound effect - User can return to Main Menu

Level
Content: - Easy button - Hard button - Home button
Action: - User can choose Easy mode or Hard mode - User can return to Main Menu

Pause
Content: - Resume button - Restart button - Sound effect bar - Music bar - Home button
Action: - User can resume the game if they paused - User can restart the game - User can adjust the volume of Sound effect and Music - User can return to Main Menu

USER FLOW



GAME ICON



This icon represents the main character Tok Bah is thinking about how to gather all the animals inside the maze to celebrate his grandson's birthday.

STORYBOARD



the main character Tok Bah awoke in the morning and realized it was a special day in the first storyline



Tok Bah recalls that it was his grandson's birthday and attempts to plan a surprise for him.

Tiba-tiba, Tok Bah ternampak kalendar yang tergantung di dinding dan teringat sesuatu. Hari ini adalah hari lahir cucunya yang bernama Anas.



Tok Bah's desire to gather all of his farm's animals to celebrate with his grandson. As a result, he must locate all of the animals that become separated in the maze.

Tok Bah ingin menjemput kawan-kawannya iaitu haiwan-haiwan di ladangnya ke majlis hari jadi cucunya.



Tok Bah is prepared to embark on the journey to rescue the animals from the maze. This is also the game's final plot

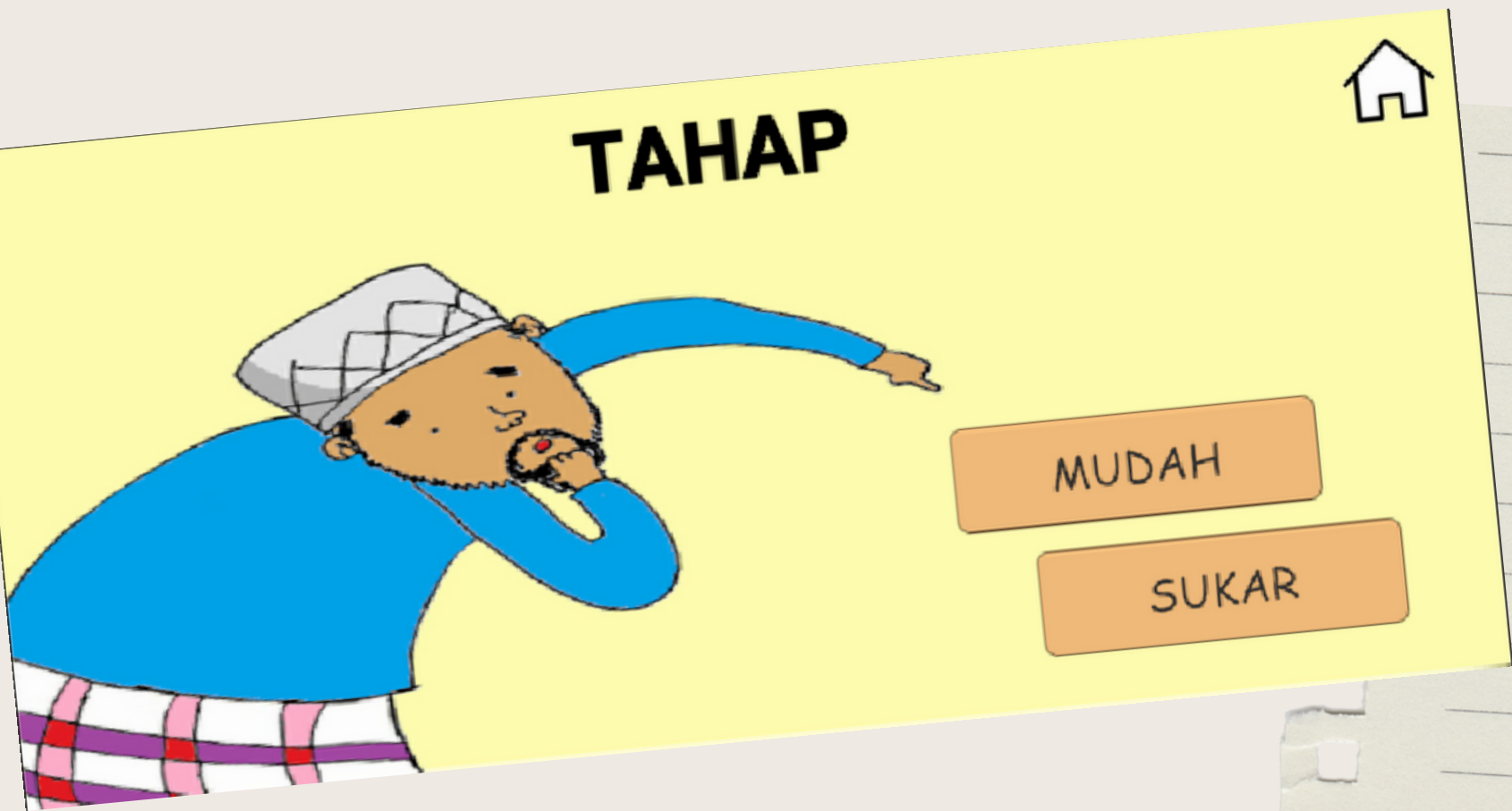
Jadi, Tok Bah memulakan perjalanan untuk mencari rakan-rakannya di ladang untuk meraikan hari jadi cucunya bersama-sama

OVERALL INTERFACE



MENU INTERFACE:
The main character "Tok Bah," who is trying to escape the maze with his friend "animals in a farm."

SCORE INTERFACE:
Best score (the shortest time it takes to complete the game) from both the easy and hard levels using this interface.



LEVEL INTERFACE:
The level interface will appear if the story line was completely displayed or if the user skipped the story line.

LEVEL COMPLETED INTERFACE:
The total time spent will be recorded and displayed to the user.



GAME OVER INTERFACE

HAPPY ENDING INTERFACE:

